

Grade for
Q2, Q3

$\frac{25}{30}$

Extra : $\frac{1}{2}$
Credit

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CHE 312

Homework 3

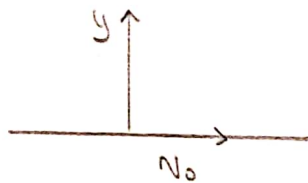
September 23, 2019

- 1) **Rayleigh problem:** Show that for a plate set suddenly in motion (the so-called Rayleigh's problem), velocity field can be described using "method of combination of variables" leading to an error function solution. (Hint: redo 4.1-1 and supply any missing steps).

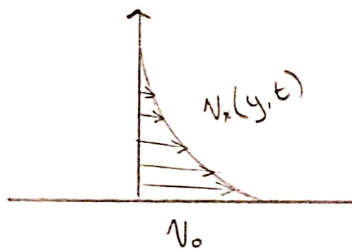
For this system $u = u(y, t)$ and $v = w = 0$



$t < 0$
Fluid @
Rest



$t = 0$
Wall set
in motion



$t > 0$
Fluid in
unsteady
flow

Continuity Equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

CONSTANT DENSITY

$$0 = 0 \checkmark$$

Equation of Motion:

Since $v = w = 0$, we will only have u term

$$\Rightarrow \rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} - \left[\frac{\partial}{\partial x} \tau_{xx} + \frac{\partial}{\partial y} \tau_{yx} + \frac{\partial}{\partial z} \tau_{zx} \right] + \rho g_x$$

$$\Rightarrow \frac{\partial u}{\partial t} = \frac{\partial}{\partial y} \tau_{yx} \quad \tau_{yx} = -\mu \frac{\partial u}{\partial y} \quad \text{where } \nu = \frac{\mu}{\rho}$$

$$\Rightarrow \frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2} \quad (4.1-1)$$

I.C. at $t \leq 0$, $u = 0$ for all y

B.C. 1) at $y = 0$, $u = u_0$ for all $t > 0$

B.C. 2) at $y = \infty$, $u = 0$ for all $t > 0$

We introduce the dimensionless velocity $\phi = \frac{u}{u_0}$

$$\frac{\partial u/u_0}{\partial t} = \nu \frac{\partial^2 u/u_0}{\partial y^2} \Rightarrow \frac{\partial \phi}{\partial t} = \nu \frac{\partial^2 \phi}{\partial y^2}$$

We know that $\phi(y, t)$ so we can plug in I.C. and boundary conditions to get: $\phi(y, 0) = 0$, $\phi(0, t) = 1$, $\phi(\infty, t) = 0$

We need to make y, t, v dimensionless so we use the following

$$\phi = \phi(\eta) \quad \eta = \frac{y}{\sqrt{4\nu t}}$$

$$\frac{d\phi}{dt} = \frac{d\phi}{d\eta} \frac{d\eta}{dt} \Rightarrow \frac{d\phi}{dt} = \frac{d\phi}{d\eta} \frac{d}{dt} \left(\frac{y}{\sqrt{4\nu t}} \right) \Rightarrow \frac{d\phi}{dt} = -\frac{1}{2} \frac{\eta}{t} \frac{d\phi}{d\eta}$$

$$\frac{d\phi}{dy} = \frac{d\phi}{d\eta} \frac{d\eta}{dy} \Rightarrow \frac{d\phi}{dy} = \frac{d\phi}{d\eta} \frac{d}{dy} \left(\frac{y}{\sqrt{4\nu t}} \right) \Rightarrow \frac{d\phi}{dy} = \frac{d\phi}{d\eta} \frac{1}{\sqrt{4\nu t}}$$

$$\frac{d^2\phi}{dy^2} = \frac{d^2\phi}{d\eta^2} \frac{d\eta^2}{dy^2} \Rightarrow \frac{d^2\phi}{dy^2} = \frac{d^2\phi}{d\eta^2} \frac{d}{dy^2} \left(\frac{y}{\sqrt{4\nu t}} \right)^2 \Rightarrow \frac{d^2\phi}{dy^2} = \frac{d^2\phi}{d\eta^2} \frac{1}{4\nu t}$$

Substitute above equations into our original equation

$$\begin{aligned} -\frac{1}{2} \frac{\eta}{t} \frac{d\phi}{d\eta} - \nu \frac{d^2\phi}{d\eta^2} \frac{1}{4\nu t} &= 0 \\ \Rightarrow +\frac{1}{2} \frac{\eta}{t} \frac{d\phi}{d\eta} + \frac{d^2\phi}{d\eta^2} \cdot 4t^{-1} &= 0 \\ \Rightarrow \frac{d^2\phi}{d\eta^2} + 2\eta \frac{d\phi}{d\eta} &= 0 \quad (\text{ODE}) \end{aligned}$$

BC for our ODE:

BC1) at $\eta = 0$, $\phi = 1$ ($N_0/N_\infty = 1$)

BC2) at $\eta = \infty$, $\phi = 0$

Now let $\frac{d\phi}{d\eta} = \psi$ and we can plug this into our original equation:

$$\frac{d\psi}{d\eta} + 2\eta \psi = 0 \Rightarrow d\psi + 2\eta \psi = \int 0 \cdot d\eta$$

$$C_1 \exp(-\eta^2) = \psi$$

$$\text{Now we get } \phi = C_1 \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta} + C_2$$

Our new dimensionless boundary conditions:

$$\text{B.C.1) } \phi = 1 \text{ @ } \eta = 0$$

$$\text{B.C.2) } \phi = 0 \text{ @ } \eta = \infty$$

} Rewritten

$$\text{Apply B.C.1} \Rightarrow 1 = C_1 \int_0^0 \exp(-\bar{\eta}^2) d\bar{\eta} + C_2 \Rightarrow \boxed{C_2 = 1}$$

$$\text{Apply B.C.2) } \Rightarrow 0 = C_1 \int_0^\infty \exp(-\bar{\eta}^2) d\bar{\eta} + 1$$

$$\underbrace{\int_0^\infty \exp(-\bar{\eta}^2) d\bar{\eta}}_{\sqrt{\pi}/2}$$

$$\Rightarrow -1 = C_1 \cdot \frac{\sqrt{\pi}}{2} \Rightarrow \boxed{C_1 = -\frac{2}{\sqrt{\pi}}}$$

Plugging in our C_1 and C_2 values to our equation we get:

$$\phi(\eta) = 1 - \frac{2}{\sqrt{\pi}} \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta}$$

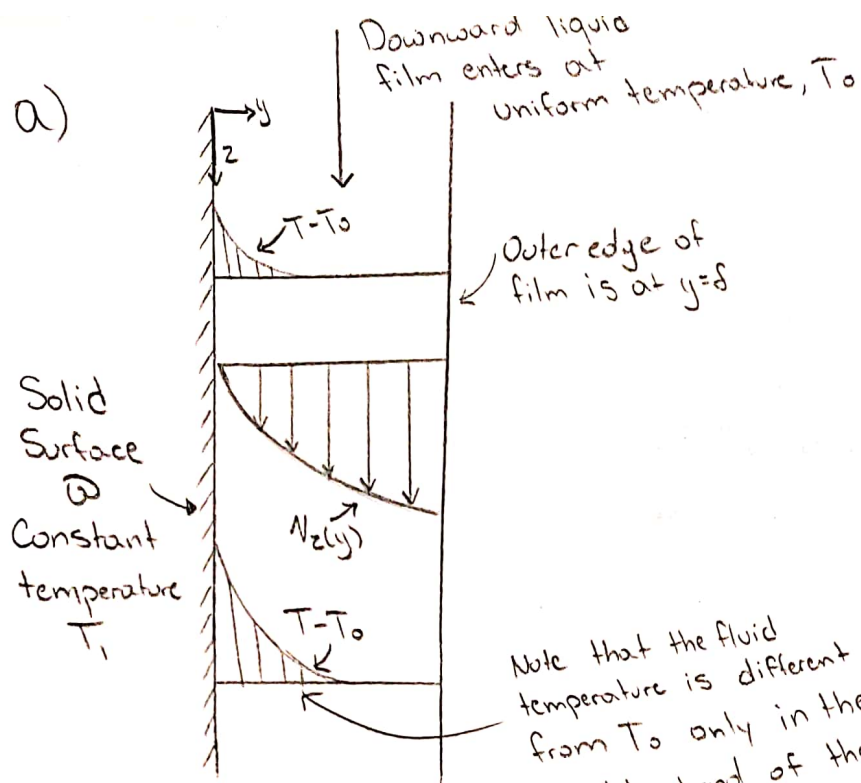
$$\phi(\eta) = 1 - \text{erf}(\eta)$$

$$\phi(\eta) = \text{erfc}$$

When you write the above equation in original variables:

$$\frac{u_x(y, t)}{u_0} = 1 - \text{erf} \frac{y}{\sqrt{4\nu t}} = \text{erfc} \frac{y}{\sqrt{4\nu t}}$$

a)



We made assumptions that:

- liquid cold fluid has considerable cooling affect on solid surface
- Short contact times that the fluid temperature changes appear only in the immediate vicinity of the wall.

We have $V_z(y)$

We will basically get equation 2.2-10 but with no $\cos\beta$ and we will have $\tau_{yz} = -\mu \frac{dV_z}{dy}$

$$\frac{d^2 V_z}{dy^2} = -\frac{\rho g}{\mu} \quad (\text{INTEGRATE})$$

$$\Rightarrow \int \frac{d^2 V_z}{dy^2} = \int -\frac{\rho g}{\mu} dy$$

$$\Rightarrow \frac{dV_z}{dy} = -\frac{\rho g y}{\mu} + C_1$$

$$\Rightarrow \int dV_z = \int -\left(\frac{\rho g y}{\mu} + C_1\right) dy$$

$$\Rightarrow V_z = -\frac{\rho g y^2}{2\mu} + C_1 y + C_2$$

BC 1) When $y=0$, $V_z=0$

BC 2) When $y=\delta$, $\frac{dv}{dy}=0$

> Plug BC 1) $0 = \frac{-\rho g(0)^2}{2\mu} + C_1(0) + C_2 \Rightarrow \boxed{C_2 = 0}$

> Plug BC 2) $\frac{dv}{dy} = \frac{-\rho g \delta^2}{2\mu} + C_1 \delta + 0$

$$\frac{\rho g \delta^2}{2\mu} = C_1 \delta \Rightarrow \boxed{C_1 = \frac{\rho g \delta}{2\mu}}$$

$$V_z = -\frac{\rho g y^2}{2\mu} + \frac{\rho g \delta y}{2\mu}$$

$$V_z = \frac{\rho g}{2\mu} \left[-\frac{y^2}{\delta^2} + \frac{\delta y}{\delta^2} \right] \Rightarrow V_z = \frac{\rho g \delta^2}{2\mu} \left[2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right]$$

When $y=\delta$, $V_z = V_{z, \max} = \frac{\rho g \delta^2}{2\mu}$

$$V_z = V_{z, \max} \left(2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right)$$

At $y=\delta$, $V = V_{z, \max} = \frac{\rho g \delta^2}{2\mu}$

$$V_z = V_{z, \max} \left(1 - \left(\frac{y}{\delta}\right)^2 \right) \Rightarrow V_{z, \max} \left(1 - \left(1 - \left(\frac{y}{\delta}\right)^2 \right) \right) \Rightarrow V_{z, \max} \left(1 - 1 + 2\left(\frac{y}{\delta}\right) + \left(\frac{y}{\delta}\right)^2 \right)$$

$$\Rightarrow V_z = 2 V_{z, \max} \left(\frac{y}{\delta}\right) = 2 \left(\frac{\rho g \delta^2}{2\mu} \right) \left(\frac{y}{\delta}\right)$$

$$\boxed{V_z \approx \frac{\rho g \delta y}{\mu}}$$

B. Assumptions:

- 1) Laminar flow, non-rippling
- 2) Heat conduction in z-direction can be neglected due to heat conduction in y-direction.

$$V_x = V_y = 0 \quad V_z(y)$$

We have from the problem statement:

$$\rho \hat{C}_p V_z \frac{dT}{dz} = k \frac{d^2 T}{dy^2}$$

The equation of motion (Appendix B.8)

$$\rho \hat{C}_p \left(\frac{dT}{dt} + V_x \frac{dT}{dx} + V_y \frac{dT}{dy} + V_z \frac{dT}{dz} \right) = - \left[\frac{dq_x}{dx} + \frac{dq_y}{dy} + \frac{dq_z}{dz} \right] - \left(\frac{d \ln p}{d \ln T} \right)_p \frac{Dp}{Dt} - (\tau \cdot \nabla)$$

$$\rho \hat{C}_p V_z \frac{dT}{dz} = - \frac{dq_y}{dy}$$

$$\text{where } q_y = -k \frac{dT}{dy}$$

$$\rho \hat{C}_p V_z \frac{dT}{dz} = k \frac{d^2 T}{dy^2}$$

$$\text{From part A } \Rightarrow V_z \approx \frac{\rho g \delta}{\mu} y$$

$$\rho \hat{C}_p \left(\frac{\rho g \delta}{\mu} y \right) \frac{dT}{dz} = k \frac{d^2 T}{dy^2}$$

$$\Rightarrow \frac{\rho^2 \hat{C}_p g \delta y}{\mu} \frac{dT}{dz} = k \frac{d^2 T}{dy^2}$$

$$\Rightarrow \frac{\rho^2 \hat{C}_p g \delta y}{\mu k} \frac{dT}{dz} = \frac{d^2 T}{dy^2}$$

$$\text{and } \beta = \frac{\mu k}{\rho^2 \hat{C}_p g \delta}$$

MULTIPLY AND DIVIDE OUT β

$$\Rightarrow y \frac{dT}{dz} = \beta \frac{d^2 T}{dy^2}$$

For short contact times we may write BC as:

B.C.1) $T = T_0$ for $z = 0$ and $y > 0$

B.C.2) $T = T_0$ for $y = \infty$ and z finite

B.C.3) $T = T_1$ for $y = 0$ and $z > 0$

→ For this one, the liquid has not come in contact with the solid yet.

→ At any distance, the temperature is the same as ambient temperature, but still not touching solid.

→ The fluid has come in contact with wall and heat penetrated a short distance of it.

Use dimensionless variables: $\theta(\eta) = \frac{T - T_0}{T_1 - T_0}$ and $\eta = \frac{y}{\sqrt[3]{9\beta z}}$
to write DE as $\frac{d^2\theta}{d\eta^2} + 3\eta^2 \frac{d\theta}{d\eta} = 0$

BC1) $\theta = 0$ when $\eta = \infty$

BC2) $\theta = 1$ when $\eta = 0$

We have: $y \frac{dT}{dz} = \beta \frac{\partial^2 T}{\partial y^2}$

IN DIMENSIONLESS FORM:

$$\eta \frac{d\theta}{d\eta} = \beta \frac{d^2\theta}{d\eta^2}$$

Lets make everything dimensionless:

$$\frac{d\theta}{dz} = \frac{d\theta}{d\eta} * \frac{d\eta}{dz} = \frac{d\theta}{d\eta} * \frac{d}{dz} \left(\frac{y}{\sqrt[3]{9\beta z}} \right) = \frac{d\theta}{d\eta} * - \frac{y}{3z\sqrt[3]{9\beta z}}$$

$$\frac{d\theta}{dy} = \frac{d\theta}{d\eta} * \frac{d\eta}{dy} = \frac{d\theta}{d\eta} * \frac{d}{dy} \left(\frac{y}{\sqrt[3]{9\beta z}} \right) = \frac{d\theta}{d\eta} * \frac{1}{\sqrt[3]{9\beta z}}$$

$$\frac{d^2\theta}{dy^2} = \frac{d}{d\eta} \left(\frac{d\theta}{d\eta} * \frac{1}{\sqrt[3]{9\beta z}} \right) \frac{d\eta}{dy} = \frac{d^2\theta}{d\eta^2} \left(\frac{1}{\sqrt[3]{9\beta z}} \right)^2$$

Plug these values into our original equation:

$$y \left(\frac{d\theta}{d\eta} * - \frac{y}{3z\sqrt[3]{9\beta z}} \right) = \beta \left(\frac{d^2\theta}{d\eta^2} \left(\frac{1}{\sqrt[3]{9\beta z}} \right)^2 \right)$$

10

$$\frac{d\theta}{d\eta} * - \frac{1}{3z} = \left(\frac{\mu k}{\rho^2 \tilde{C}_p g \delta} \right) \frac{d^2\theta}{d\eta^2} \left(\frac{1}{\sqrt[3]{9\beta z}} \right)$$

B.C.1) when $\eta=0$, $\theta=1$

B.C.2) when $\eta=\infty$, $\theta=0$

If we use our dimensionless quantities: $\theta(\eta) = \frac{T-T_0}{T_1-T_0}$ & $\eta = \frac{y}{\sqrt[3]{9\beta z}}$

We will get an equation that i've practiced

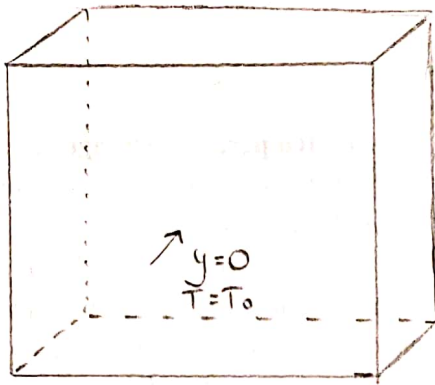
Showing alot: $\theta = 1 - \frac{2}{\sqrt{\pi}} \int_0^\eta \exp(-\tilde{u}^3) d\tilde{u} + 1$

We get the following ODE:

$$\frac{d^2\theta}{d\eta^2} + 3\eta^2 \frac{d\theta}{d\eta} = 0$$

a.

LIQUID
CRYSTALLINE
POLYMER
(Changes color
with temp)



- We want to know how fast color changes and how we can control it
- provide constant heat flux (q_0) some with heater or party

Assumptions:

- Initial temp is T_0
- Constant heat flux appears @ $t=0$ at the wall i.e. $y=0$
- Wall is very thick that you can say its semi-infinite
- Temp profiles $T(y, t)$ for short times.
- Solid wall occupying space from $y=0$ to $y=\infty$
- Assume k is independent of temp & position

For solid's, the energy equation when combined with Fourier's Law of heat conduction $\Rightarrow \rho \hat{C}_p \frac{dT}{dt} = (\nabla \cdot k \nabla T)$

When k is independent of temp & position: $\frac{dT}{dt} = \alpha \nabla^2 T$

Where $\alpha = \frac{k}{\rho \hat{C}_p}$

For one-dimensional heat conduction (12.1-33) $\Rightarrow \frac{dT}{dt} = \alpha \frac{d^2 T}{dy^2}$

Recognizing that we can implement Fourier's law into the above by multiplying by $-k$ and d/dy , we get:

$$\frac{d}{dt} T = \alpha \frac{d^2}{dy^2} T \Rightarrow \frac{d}{dt} \left(\underset{\downarrow q_{y1}}{-k \frac{dT}{dy}} \right) = \alpha \frac{d^2}{dy^2} \left(\underset{\downarrow q_{y1}}{-k \frac{dT}{dy}} \right) \Rightarrow \frac{dq_{y1}}{dt} = \alpha \frac{d^2 q_{y1}}{dy^2} \quad (\text{Equ. 12.1-35})$$

c.) when $t \leq 0$, $V_x = 0$ for all y

B.C. 1) @ $y=0$, $q_y = q_0 \mathcal{R}\{e^{i\omega t}\}$

B.C. 2) @ $y=\infty$, $q_y = 0$

We can introduce a dimensionless heat flux and express it as:

$$\theta = \frac{q_y}{q_0}$$

$$\frac{dq_y/q_0}{dt} = \alpha \frac{d^2 q_y/q_0}{dy^2} \Rightarrow \frac{d\theta}{dt} = \alpha \frac{d^2 \theta}{dy^2}$$

With $\theta(y, 0) = 0$, $\theta(0, t) = 1$, and $\theta(\infty, t) = 0$

Also, for our equation above to be all dimensionless, we need to introduce this expression: $\theta = \theta(\eta)$ where $\eta = \frac{y}{\sqrt{4\alpha t}}$ (Chapter 4)

Change PDE into ODE:

$$\frac{d\theta}{dt} = \frac{d\theta}{d\eta} \cdot \frac{d\eta}{dt} \Rightarrow \frac{d\theta}{d\eta} \cdot \frac{d}{dt} \left(\frac{y}{\sqrt{4\alpha t}} \right) \Rightarrow \frac{d\theta}{dt} = \frac{d\theta}{d\eta} \left(-\frac{1}{2} \frac{\eta}{t} \right)$$

$$\frac{d\theta}{dy} = \frac{d\theta}{d\eta} \cdot \frac{d\eta}{dy} \Rightarrow \frac{d\theta}{d\eta} \cdot \frac{d}{dy} \left(\frac{y}{\sqrt{4\alpha t}} \right) \Rightarrow \frac{d\theta}{dy} = \frac{d\theta}{d\eta} \left(\frac{1}{\sqrt{4\alpha t}} \right)$$

$$\frac{d^2 \theta}{dy^2} = \frac{d^2 \theta}{d\eta^2} \cdot \frac{d\eta^2}{dy^2} \Rightarrow \frac{d^2 \theta}{d\eta^2} \cdot \frac{d}{dy^2} \left(\frac{y}{\sqrt{4\alpha t}} \right)^2 \Rightarrow \frac{d^2 \theta}{dy^2} = \frac{d^2 \theta}{d\eta^2} \left(\frac{1}{4\alpha t} \right)$$

Substitute above equations into our original PDE:

$$\frac{d\theta}{dt} = \alpha \frac{d^2 \theta}{dy^2} \Rightarrow \frac{d\theta}{d\eta} \left(-\frac{1}{2} \frac{\eta}{t} \right) = \alpha \frac{d^2 \theta}{d\eta^2} \left(\frac{1}{4\alpha t} \right) \Rightarrow -2\eta \frac{d\theta}{d\eta} = \frac{d^2 \theta}{d\eta^2}$$

$$\frac{d^2 \theta}{d\eta^2} + 2\eta \left(\frac{d\theta}{d\eta} \right) = 0$$

The new boundary conditions for dimensionless form are:

B.C. 1) $\eta = 0$ when $\theta = 1$

B.C. 2) $\eta = \infty$ when $\theta = 0$

Now let $\frac{d\theta}{d\eta} = \psi$ so we can get first ODE

$$\psi = \frac{d\theta}{d\eta} = C_1 \exp(-\eta^2) \quad (\text{INTEGRATE AGAIN})$$

$$\theta(\eta) = C_1 \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta} + C_2$$

Apply BC 1) $\Rightarrow 1 = C_1 \int_0^0 \exp(-\bar{\eta}^2) d\bar{\eta} + C_2 \Rightarrow C_2 = 1$

Apply BC 2) $\Rightarrow 0 = C_1 \underbrace{\int_0^\infty \exp(-\bar{\eta}^2) d\bar{\eta}}_{\sqrt{\pi}/2} + 1 \Rightarrow C_1 = -2/\sqrt{\pi}$

Now plug in B.C.

$$\theta(\eta) = \underbrace{-\frac{2}{\sqrt{\pi}} \int_0^\eta \exp(-\bar{\eta}^2) d\bar{\eta}}_{\text{erf}} + 1$$

$$\Rightarrow \theta(\eta) = 1 - \text{erf}(\eta)$$

$$\theta(\eta) = 1 - \text{erf}\left(\frac{y}{\sqrt{4\alpha\tau}}\right)$$

OR $\theta(\eta) = \text{erfc}(\eta)$

we will start getting rid of dimensionless quantities:

$$\frac{q_y}{q_0} = \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right)$$

$$\Rightarrow q_y = q_0 \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right) \quad \left(\text{Fourier's Law } q_y = -k \frac{dT}{dy}\right)$$

$$\Rightarrow -k \frac{dT}{dy} = q_0 \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right)$$

$$\Rightarrow \int_{T_0}^T dT = -\frac{q_0}{k} \int_y^{\infty} \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right) dy$$

$$\Rightarrow T - T_0 = -\frac{q_0}{k} \int_y^{\infty} \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right) dy$$

Basic Integral: $\int \operatorname{erfc}(ax) dx = x \operatorname{erfc}(ax) - \frac{1}{a\sqrt{\pi}} \exp(-a^2 x^2)$

(integral of product of error functions with other functions)

Now by solving the integral:

$$T - T_0 = \frac{q_0}{k} \left(\bar{y} \operatorname{erfc}(\bar{\eta} \bar{y}) - \frac{1}{\bar{\eta} \sqrt{\pi}} \exp(-\bar{\eta}^2 \bar{y}^2) \right) \Big|_y^{\infty}$$

$$T - T_0 = \frac{q_0}{k} \left(\bar{y} \operatorname{erfc}(\infty) - \sqrt{\frac{4\alpha t}{\pi}} \cdot \frac{1}{e^{\infty}} \right) - \left(\bar{y} \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right) - \sqrt{\frac{4\alpha t}{\pi}} \cdot \exp\left(\frac{-y^2}{4\alpha t}\right) \right)$$

$$T - T_0 = -\bar{y} \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right) + \frac{\sqrt{4\alpha t}}{\sqrt{\pi}} \exp\left(\frac{-y^2}{4\alpha t}\right)$$

b. We need to show that it satisfies the 1D heat conduction equation for temperature, and that it satisfies I.C. + B.C.

$$\frac{dT}{dt} = \alpha \frac{d^2T}{dy^2} \quad (\text{SHOW THAT IT SATISFIES LARGE } y \text{ AND})$$

B.C. 1) When $y=0$, $q_y = q_0$

$$\Rightarrow q_y = q_0 \operatorname{erfc}(n)$$

$$\Rightarrow q_y = q_0 \operatorname{erfc}\left(\frac{0}{\sqrt{4\alpha t}}\right) \Rightarrow q_y = q_0 \checkmark$$

I.C.) When $q_y = 0$ then $t=0$

$$q_y = q_0 \operatorname{erfc}\left(\frac{y}{\sqrt{4\alpha t}}\right)$$

$$q_y = q_0 \operatorname{erfc}\left(\frac{y}{0} \rightarrow \infty\right) \Rightarrow q_y = q_0 \operatorname{erfc}(\infty) \Rightarrow q_y = 0 \checkmark$$

B.C. 2) When $y=\infty$, then $T=T_0$

$$T - T_0 = \frac{q_0}{k} \left(-y \operatorname{erfc}(\infty) + \frac{\sqrt{4\alpha t}}{\sqrt{\pi}} * \exp(\infty) \right)$$

$$T - T_0 = 0 \Rightarrow T = T_0 \checkmark$$

The initial and boundary conditions satisfy our equation so Hippo should apologize.